Temporal trajectories of type 2 diabetes complications from inpatient data collected from 2006 to 2021 at the Centre hospitalier universitaire de Sherbrooke (CHUS)

Noé Carème Fouotsa Manfouo, Ruth Ndjaboue, Gerard Ngueta

[a] Centre de recherche du CHU de Sherbrooke, Québec · [b] Centre de recherche sur le vieillissement, Sherbrooke, Québec

INTRODUCTION

Type 2 diabetes mellitus (T2DM) leads to long-term complications — nephropathy, retinopathy, neuropathy, vascular diseases and amputations — that develop **sequentially and often irreversibly**. T2DM prevalence keeps rising in North America, including Canada.

Few studies model trajectories of **more than two complications** or compare them across **age and sex** in a Canadian population.

OBJECTIVES

1. Build time-ordered trajectories of **2 to 5 complications**: nephropathy (E11.2), retinopathy (E11.3), neurologic diseases (E11.4), vascular diseases (E11.5) and amputations (L97).
2. Examine **age- and sex-specific** patterns and compare them with similar studies.

METHODS

Data — CHUS inpatient records, 2006–2021, patients with T2DM

▼

Sequences — each patient's complications (ICD-10 E11.2–E11.5, L97) ordered by date of first diagnosis

▼

Stratification — sex (F / M) × age at diagnosis (< 50 / ≥ 50 years)

▼

Conditional probabilities
k-dimensional matrices M_k, k = 2...5

$$P(x_k \mid x_1 \ldots x_{k-1}) = M_k[x_1 \ldots x_k] \;/\; \Sigma_j \, M_k[x_1 \ldots x_{k-1}, j]$$

PrefixSpan
frequent sequential patterns (support ≥ min_sup)

$$\text{Likelihood}(X) = \text{Sup}(X) \;/\; \Sigma_i \, \text{Sup}(X_i) \times 100$$

▼

Output — most likely trajectory for each length (2–5) and sub-group

CONCLUSION

- **Novel contribution** — one of the first models of up to five T2DM complication trajectories using Canadian inpatient data.
- **Demographics matter** — age- and sex-specific progression patterns should inform T2DM management.
- **Act early** — early, comprehensive and tailored management can prevent or delay complications.
- **Next steps** — integrate more diverse datasets and additional clinical factors.

RESULTS — MOST LIKELY COMPLICATION TRAJECTORIES

Nephropathy E11.2 · Retinopathy E11.3 · Neuropathy E11.4 · Vascular diseases E11.5 · Amputation L97

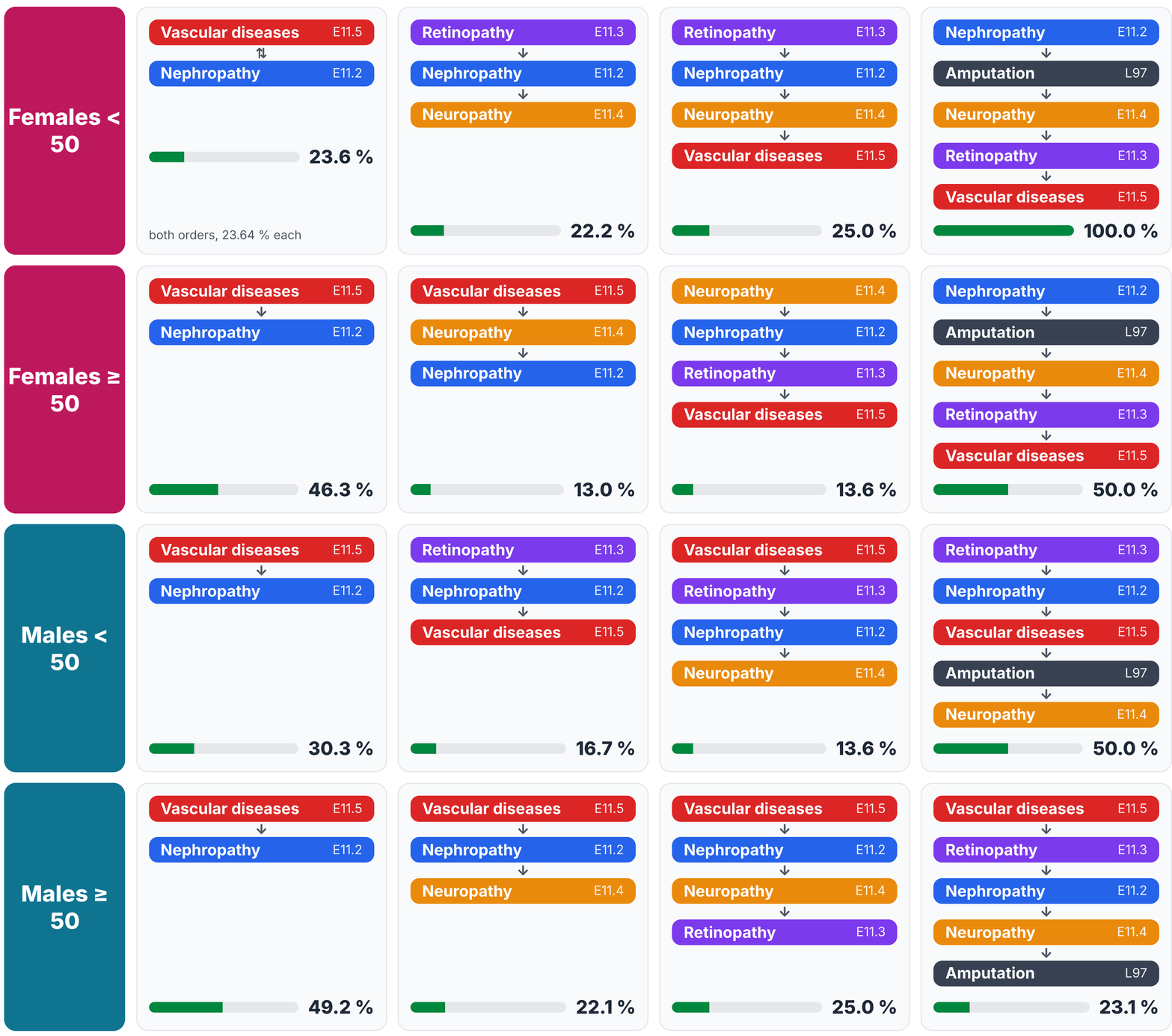


Fig. 1. Most likely trajectory of 2, 3, 4 and 5 complications for each sub-group (PrefixSpan). Bars show the likelihood, i.e. the share of support of the sequence among frequent sequences of the same length in that sub-group. Redrawn from the original trajectory graph.

KEY FINDINGS

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sub-groups share the same leading 2-step trajectory: **vascular diseases → nephropathy** (23.6 % to 49.2 %), confirmed by both methods.

≥ 50 y
Older patients concentrate on this pathway: **49.2 %** (men) and **46.3 %** (women) vs 30.3 % and 23.6 % before 50.

5 steps
Long trajectories involve **neuropathy and amputation**; in men ≥ 50 amputation is the final stage after neuropathy.

DISCUSSION

- **Consistent patterns** — vascular diseases followed by nephropathy is the most common start, with both conditional probabilities and PrefixSpan.
- **Age & sex differences** — trajectories of 3–5 complications differ markedly between sub-groups.
- **Advanced sequences** — neuropathy precedes severe outcomes such as amputation.
- **Literature** — the frequent nephropathy–neuropathy association agrees with prior work; PrefixSpan aligned best with published trajectories.

REFERENCES

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3. Jensen AB, Moseley PL, Oprea TI, et al. Temporal disease trajectories condensed from population-wide registry data covering 6.2 million patients. *Nat Commun*. 2014;5:4022.

Explore the method interactively
Web demo of the trajectory analysis (simulated data, no patient information).

Contact: Noé Carème Fouotsa Manfouo · github.com/noe2019

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